SENIOR GAME ENGINEER · UE5 / C++

Siddhant Bhati

Gameplay, combat, networking & AI systems

Delhi, India · UTC+5:30
+91 89207 12857
siddhantbhati@gmail.com
linkedin.com/in/siddhant-bhati-100a44b2
skiggs.github.io/website

AT A GLANCE

7+ years
shipping C++ game code since 2019

Shipped
Project Motor Racing on PC, PS5 & Xbox Series X|S

Rollback
netcode combat architecture, built from scratch in UE5

CORE SKILLS

Languages
C++ (STL, Boost), Python

Engines
Unreal Engine 4/5, GIANTS Engine, Lumberyard, Unity, SFML, DirectX

Gameplay & combat
Movement components, animation state machines, input buffering, ability frameworks, cameras, replays

Networking
Rollback netcode, deterministic fixed-step sim, dedicated servers, UE replication

AI
Behavior Trees, Blackboard, EQS, state machines, A*, squad bots

Engineering
System design, DSA, game math, data-oriented design, profiling, GoogleTest

EDUCATION

B.Sc. (Hons) Computer Science
Bhaskaracharya College of Applied Sciences, University of Delhi · 2015–2018

PROFILE

C++ game engineer across Unreal Engine, GIANTS Engine and Lumberyard. Shipped **Project Motor Racing**, owning its replay, camera, penalty and pit systems; built battle-royale bot AI at Mayhem Studios. Now lead engineer on **Tilted Slammers**, a UE5 tactical arena brawler. Open to remote contract, part-time and full-time roles.

EXPERIENCE

Founder & Lead Engineer
2024 – Present

Firetail Studios LLP · Delhi · *Tilted Slammers* (UE5 C++)

- Architected the combat stack: fixed-step deterministic simulation with client-side rollback netcode on server-authoritative dedicated servers.
- Built a custom movement component, phase-based animation state machine and 9-frame input buffer with gesture history, modelled on platform fighters.
- Designed an ability framework (staged, multi-charge, placeable) and a team-owned arena wall system.
- Profiled the packaged server: 4-player sim step ~0.7 ms avg, p99 < 2.7 ms at 150 ms latency and 5% loss; ~10 matches per CPU core.
- Lead a freelance art and animation team. Selected for IGDC I/P Connect 2026.

Gameplay Programmer
Shipped Nov 2025

Straight4 Studios · Remote · *Project Motor Racing*

`PC · PS5 · XBOX SERIES X|S`

- Developed core race systems in Unreal Engine 5, then carried them through the studio's migration to GIANTS Engine.
- Built the replay system and race camera systems, including trackside broadcast cameras.
- Implemented the race penalty system and trackside pit systems.

C++ Unreal Engine Developer
Jun 2022 – Jul 2023

Mayhem Studios · Bangalore · *Underworld Gang Wars* (battle royale)

- Built the bot AI: cover selection, shooting from cover, revive, and integration with the character weapon system.
- Drove bot behaviour with a hybrid of Behavior Trees and state machines, using EQS for positional queries.
- Coordinated bot behaviour across duo and squad modes.
- Developed backend features and GoogleTest coverage.

Siddhant Bhati

SENIOR GAME ENGINEER · UE5 / C++

OPEN TO

Remote contract, part-time or full-time

Gameplay, combat, AI and networking systems in UE5 C++.

Based in UTC+5:30: full overlap with Europe, partial with US hours.

RECOGNITION

IGDC I/P Connect 2026
Tilted Slammers selected

gamescom asia 2026
GameDev India scholar, Bangkok

CONTACT

+91 89207 12857
siddhantbhati@gmail.com
LinkedIn profile
Portfolio

EXPERIENCE, CONTINUED

C++ Unreal Engine Developer
Sep 2021 – May 2022

Freelance · Remote

- Set up the backend for an RTS game.
- Built core AI for a MOBA with Behavior Trees and Blackboard, and implemented character spells.

Game Engineer (C++)
Jul 2020 – Sep 2021

Pragmatic Play · Noida · *Fantastic League 2021*

- Reverse-engineered the 2005 legacy codebase and ported it to Lumberyard; migrated the engine from VS2005 to VS2017 for debugging.
- Wrote a bytecode reader to parse and organise the legacy binary data.
- Built a custom Lumberyard Gem for player animation with custom blending and name-driven motion switching.
- Reimplemented the legacy ball physics in Lumberyard.

C++ Game Developer
Feb 2019 – May 2020

Ingenuity Gaming · Noida

- Built core parts of a slot-game engine for casino cabinets, using state machines to drive game math and presentation flow.
- Created an XML-driven module that automated on-screen animation setup, and a data-management interface for game configuration.

PERSONAL PROJECTS

Hotlike Miami
C++ · SFML

Top-down game built from scratch: game loop, asset manager, state machine, text-file levels, collision math, A* and a data-oriented message bus. Hitman-inspired enemy AI takes cover, revives allies, dodges grenades and judges whether a weapon pickup is worth the risk.

Collision optimisation
C++ · SIMD

Real-time collision for 6,000+ shapes using quadtrees, spatial hashing, an entity component system and SIMD.

Unreal Engine experiments
UE4 / UE5

Procedural-animation character controller with IK and Control Rig; NPC AI on navmesh; replication and multicast samples; a simple ML agent in C++.